

EBU6505 – Reasoning and Agents

Block 2, Day 2 – Exact Inference in Bayesian Networks

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Teaching Week 4

What have we covered this week?

- Day 1: what a Bayesian network is
- nodes, arrows, and CPTs
- how the graph encodes conditional independence
- how a network represents one joint probability model

Main idea: Yesterday we learned how to read the model. Today we learn how to get an answer from the model.

What are we going to cover today?

- what an inference query looks like
- enumeration as the direct exact method
- variable elimination as the smarter exact method
- why elimination order and treewidth matter
- when exact inference becomes too expensive

Main idea: The whole lecture answers one question: once we have a Bayesian network, how do we use it to compute a probability?

- *Artificial Intelligence: A Modern Approach*, Russell and Norvig (4th ed.)
- AIMA 4e, Chapter 13: Exact Inference in Bayesian Networks
- *Artificial Intelligence: Foundations of Computational Agents*, David Poole and Alan Mackworth (3rd ed.)
- Poole 3e, Chapter 9: Probabilistic Inference

Main idea: Both books cover the same two exact methods for today: enumeration and variable elimination.

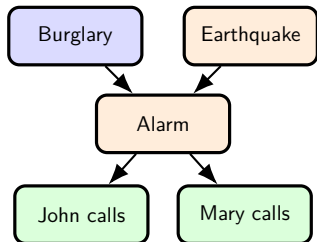
Learning Outcomes for Today

- identify the query, evidence, and hidden variables in a BN question
- explain enumeration in simple words
- explain how variable elimination avoids repeated work
- read the meaning of factor product, summing out, and normalisation
- explain why exact inference gets harder in more connected graphs

Main idea: If these five ideas are clear, students can follow exact-inference calculations without getting lost in notation.

Warm-Up Question

Question: If John and Mary both call, how do we compute the probability of a burglary?



What is known?

We know the graph and the CPTs.
We observe two calls.

What is unknown?

We want the probability of burglary after seeing the calls.

$$P(B \mid J=t, M=t)$$

Main idea: Inference means: use the model plus the evidence to compute the probability of the query

How Do We Read an Inference Query?

Question: Which part is the answer, and which part is the evidence?

$$P(B \mid J=t, M=t)$$

Query

B
the variable we want to
know about

Evidence

$J=t, M=t$
the facts we have ob-
served

Hidden variables

A, E
not observed, but still
relevant

Main idea: A Bayesian-network query always separates into query variables, observed evidence, and hidden variables that must be handled in the calculation.

Exact or Approximate?

Question: What kind of answer do we want today?

Exact inference

- returns the exact probability
- uses full probability calculation
- good for small and moderate networks

Approximate inference

- returns an estimate
- often uses sampling
- helpful when exact inference is too expensive

Main idea: Today we focus on exact inference, so every answer comes from full probability calculation rather than estimation.

General Goal of Exact Inference

Question: What does the exact-inference formula say in plain words?

$$P(X \mid e) = \alpha \sum_y P(X, e, y)$$

In simple words

Keep the query variable.
Hold the evidence fixed.
Add over every hidden possibility.
Then normalise.

Symbols used on this slide

X : query variable

e : observed evidence

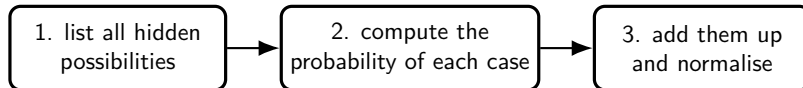
y : hidden variables

α : normalisation constant

Main idea: Exact inference is just controlled summation over the hidden parts of the model.

Enumeration: The Most Direct Method

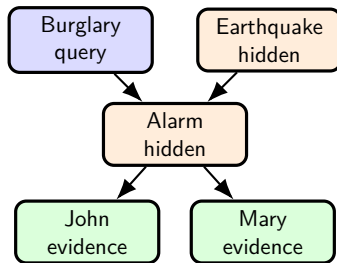
Question: If we do inference in the simplest possible way, what do we do?



Main idea: Enumeration is correct and easy to explain, but it can become slow because the number of hidden cases grows quickly.

Which Variables Are Hidden in the Alarm Query?

Question: When we ask about burglary, which variables do we still have to sum over?



Main idea: For this query, the evidence is fixed, the query is Burglary, and the hidden variables are Earthquake and Alarm.

Write the Enumeration Query

Question: What exact expression should we compute for the alarm example?

$$P(B \mid j, m) = \alpha \sum_e \sum_a P(B, e, a, j, m)$$

Why two sums?

Earthquake and Alarm are hidden, so we must add over both.

What stays fixed?

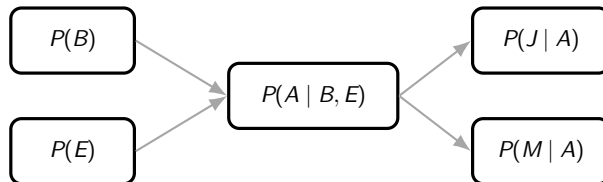
The query variable B and the evidence j, m stay in the expression.

Main idea: Enumeration writes the full query, then sums over the hidden variables.

Break the Joint into Local Pieces

Question: Where does the graph help in the calculation?

$$P(B, e, a, j, m) = P(B) P(E) P(A \mid B, E) P(J \mid A) P(M \mid A)$$



Main idea: The graph lets us replace one big joint probability with several small local terms.

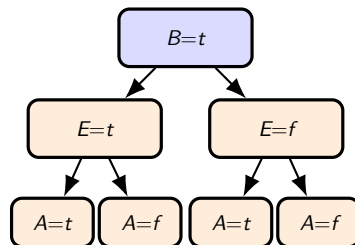
Enumeration Case 1: Assume Burglary Is True

Question: What do we do after choosing one value of the query variable?

$$\text{score}(B=t) = \sum_e \sum_a P(B=t, e, a, j, m)$$

Hidden cases to add

$(E=t, A=t)$, $(E=t, A=f)$,
 $(E=f, A=t)$, $(E=f, A=f)$



Main idea: For one query value, enumeration adds the probability of every compatible hidden world.

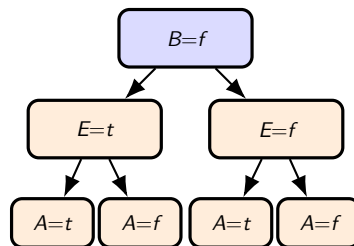
Enumeration Case 2: Assume Burglary Is False

Question: Why do we need a second case for the query variable?

$$\text{score}(B=f) = \sum_e \sum_a P(B=f, e, a, j, m)$$

Why repeat the process?

The final answer is a distribution over B , so we need one score for $B=t$ and one score for $B=f$.



Main idea: Exact inference returns a full normalized distribution, not just one isolated number.

Normalise the Two Scores

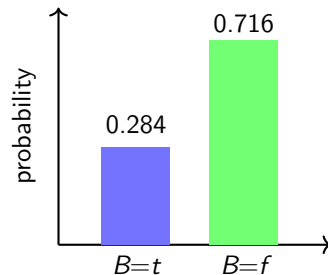
Question: How do two unnormalized scores become a valid probability distribution?

$$P(B \mid j, m) = \frac{\text{score}(B)}{\sum_B \text{score}(B)}$$

Result for the alarm example

$$P(B=t \mid j, m) \approx 0.284$$

$$P(B=f \mid j, m) \approx 0.716$$



Main idea: Normalisation rescales the scores so they add up to 1 and become a proper probability distribution.

In-Class Checkpoint

Question: If we observe John and Mary, which variables are evidence and which are hidden?

Discuss with a partner

- identify the query variable
- identify the evidence variables
- identify the hidden variables
- explain why enumeration needs summation

Expected answer

Query: B

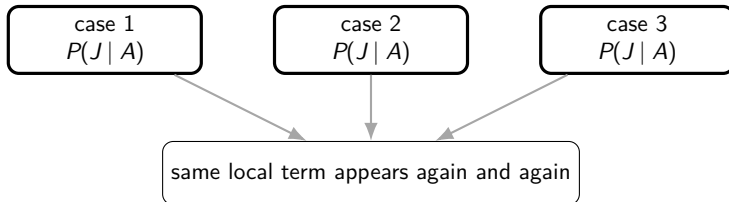
Evidence: J, M

Hidden: A, E

Main idea: Students should be able to say what is known, what is unknown, and what must be summed over before moving on.

Why Is Enumeration Wasteful?

Question: Where does enumeration waste time?

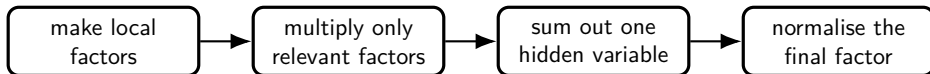


Enumeration recomputes many of the same local pieces in different hidden cases.

Main idea: Variable elimination is useful because it stores shared intermediate work instead of repeating it.

Variable Elimination: The Smarter Exact Method

Question: How can we stay exact but avoid repeated work?



Main idea: Variable elimination is still exact, but it organises the work so common pieces are reused.

What Is a Factor?

Question: What object does variable elimination work with?

A	$P(J=t \mid A)$
t	0.90
f	0.05

Simple meaning

A factor is just a small table that stores numbers for some variables.

Example

After applying evidence, a CPT can become a factor over fewer variables.

Main idea: Variable elimination works with small tables called factors, not with one giant joint table.

Two Operations Drive Variable Elimination

Question: What are the only two operations we really need?

1. Pointwise product

line up matching rows and multiply the numbers

$$f(A, B) \times g(B, C)$$

2. Summing out

remove a hidden variable by adding over its values

$$\sum_A h(A, B, E)$$

Main idea: VE keeps combining useful factors and then removing hidden variables one at a time.

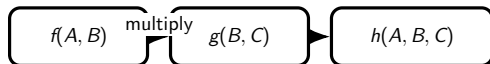
What Does Pointwise Product Mean?

Question: How do two small factors become one larger factor?

$$f(A, B) \times g(B, C) = h(A, B, C)$$

How to think about it

Keep the shared variable aligned. Multiply row by row. The result remembers all variables that still matter.

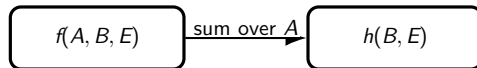


Main idea: Pointwise product combines local information into one shared factor.

What Does Summing Out Mean?

Question: How do we remove a hidden variable after using it?

$$h(B, E) = \sum_A f(A, B, E)$$



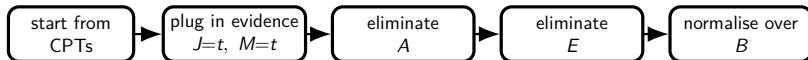
Plain meaning

After variable A has done its job, add over its values and keep only the variables we still need.

Main idea: Summing out removes one hidden variable while preserving its overall effect on the remaining variables.

Variable Elimination Workflow for the Alarm Query

Question: What is the overall plan for this query?



Main idea: VE chooses an order for the hidden variables, removes them one by one, and leaves a final factor for the query variable.

Step 1: Build the Initial Factors

Question: What factors do we start with for the alarm network?

$$f_1(B) = P(B)$$

$$f_2(E) = P(E)$$

$$f_3(A, B, E) = P(A \mid B, E)$$

$$f_4(J, A) = P(J \mid A)$$

$$f_5(M, A) = P(M \mid A)$$

Each factor comes directly from one local CPT in the network.

Main idea: The graph tells us exactly which local factors exist before any evidence is applied.

Step 2: Apply the Evidence

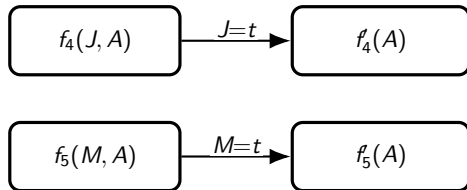
Question: How do the call observations simplify the factors?

$$J=t, \quad M=t$$

$$f'_4(A) = P(J=t \mid A), \quad f'_5(A) = P(M=t \mid A)$$

What changed?

The observed variables disappear from the factor arguments because their values are now fixed.



Main idea: Evidence reduces the size of some factors before elimination begins.

Step 3: Eliminate Alarm

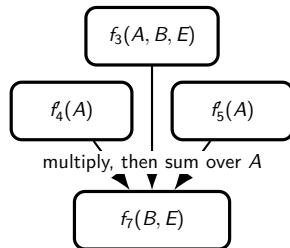
Question: What do we do with the first hidden variable?

$$f_6(A, B, E) = f_3(A, B, E) f_4(A) f_5(A)$$

$$f_7(B, E) = \sum_A f_6(A, B, E)$$

Meaning

Multiply the three factors that mention A .
Then sum over A to remove it.



Main idea: After using Alarm, keep a smaller factor over B and E .

Step 4: Eliminate Earthquake

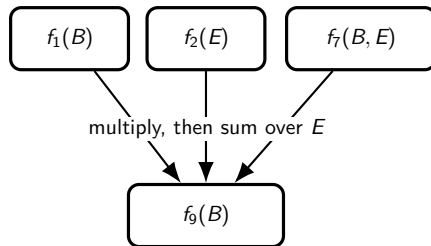
Question: What is left after Alarm has been removed?

$$f_8(B, E) = f_1(B) f_2(E) f_7(B, E)$$

$$f_9(B) = \sum_E f_8(B, E)$$

What remains now?

Only the query variable B remains, so we are almost finished.



Main idea: VE removes the hidden variables one by one until only the query variable is left.

Step 5: Normalise the Final Factor

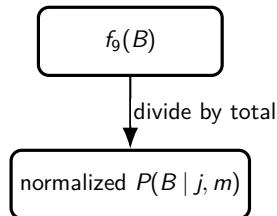
Question: Why is the last factor not yet the final answer?

$$P(B \mid J=t, M=t) = \frac{f_9(B)}{\sum_B f_9(B)}$$

Final answer

$$P(B=t \mid j, m) \approx 0.284$$

$$P(B=f \mid j, m) \approx 0.716$$



Main idea: Normalisation is the last step in both enumeration and variable elimination.

Why Is Variable Elimination Better Than Enumeration?

Question: What work does VE save?

Enumeration

- lists many hidden cases
- repeats local calculations
- conceptually simple

Variable elimination

- stores intermediate factors
- reuses shared work
- often much faster

Main idea: Both methods are exact, but VE is usually preferred because it avoids repeated computation.

Does Elimination Order Matter?

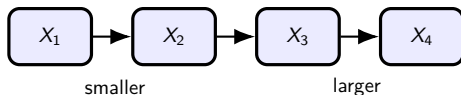
Question: Can the same query become easier or harder depending on the order?

Good order

remove a lower-impact variable first
smaller factors

Bad order

remove a highly connected variable
early
larger factors



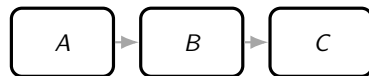
Main idea: Elimination order matters because it changes factor size and cost.

What Is Treewidth in Simple Words?

Question: What tells us whether exact inference will stay manageable?

Simple intuition

Treewidth asks: what is the biggest group of variables we must keep together inside one intermediate factor?



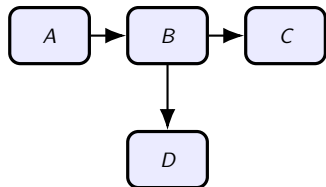
largest group here has 3 variables

$\text{treewidth} \approx \text{largest group size} - 1$

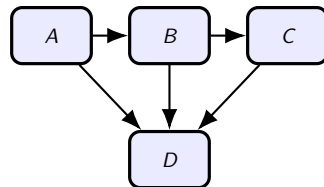
Main idea: Exact inference is easier when the biggest factor stays small.

Sparse Graph or Dense Graph?

Question: Which kind of network usually gives easier exact inference?



sparser graph
smaller factors

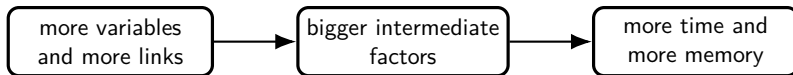


denser graph
larger factors

Main idea: More connections usually mean larger factors and harder exact inference.

When Does Exact Inference Become Hard?

Question: Why can exact inference become too expensive in bigger models?



This is why exact inference works well on some networks, but not on every realistic large-scale model.

Main idea: The challenge is not the definition of exact inference. The challenge is the size of the intermediate factors it creates.

Where Are We Going Next?

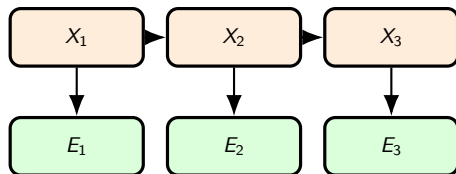
Question: What changes when uncertainty unfolds over time instead of one static snapshot?

Today

one Bayesian network
one fixed inference query

Next lecture

repeated uncertain states across time
state today affects state tomorrow



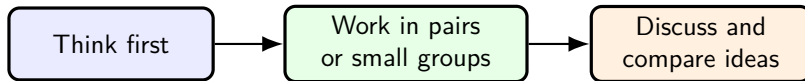
Main idea: Day 3 keeps probabilistic inference but adds time and sequential evidence.

Key Takeaways

- exact inference answers a query using the model plus evidence
- enumeration is the direct method: list, multiply, add, normalise
- variable elimination is smarter because it reuses intermediate factors
- elimination order matters because it changes factor size
- treewidth gives an intuition for why exact inference can become hard

Main idea: Students should now see exact inference as a structured workflow, not as a disconnected set of formulas.

In-Class Exercises



Use this time to work on today's in-class tasks and prepare one clear answer you can discuss.

Main idea: Use the exercise time to explain the inference workflow in clear everyday language, not only in formulas.

Further Study

- Read Poole Chapter 9 and AIMA Chapter 13.
- Practice: Poole Ch.9, Exercises 9.8, 9.9, 9.11, and 9.15.
- AIMA Bayesian-network exercise page: 14.18, 14.19, and 14.21.
- Focus on enumeration, variable elimination, and why elimination order affects difficulty.